ArogyaWatch UP

Remote monitoring and early risk detection. Demo with simulated patients.

Doctor dashboard | Patient view | AI chat | My health | Care hub

▶ Run live demo | Village network: online | Pause

NEWS2-based risk score | AI trend forecast + LLM copilot | z-score and χ^2 anomaly checks | Works offline, SMS fallback | AI triage chat | Daily tracker + insights | Emergency help | Run tracker + daily bio-data | SQL patient records | E-prescription + medical store | Hindi voice for patients | ASHA + eSanjeevani workflow

2	0	5	0
High risk	Need review	Stable	Predicted to worsen

Live

Network online, readings arriving

PATIENTS BY URGENCY

Mohd. Irfan — 10 High risk
55 · Sandila, Hardoi
HR 126 BP 108/66 SpO_2 87 38.5° 106

Savitri Devi — 7 High risk
72 · Misrikh, Sitapur
HR 103 BP 123/79 SpO_2 95 37.3° 311

Phoolmati Devi — 1 Stable
74 · Payagpur, Bahraich
HR 86 BP 142/88 SpO_2 94 37° 130
Manual entry

Geeta Devi — 0 Stable
48 · Maharajganj, Raebareli
HR 74 BP 117/74 SpO_2 98 36.7° 94

Kamla Verma — 0 Stable
63 · Hasanganj, Unnao
HR 71 BP 121/74 SpO_2 97 36.6° 101

Ramesh Yadav — 0 Stable
68 · Siddhaur, Barabanki
HR 78 BP 132/85 SpO_2 96 36.7° 122

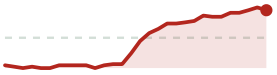
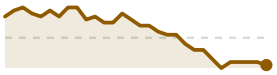
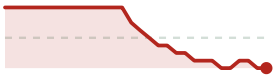
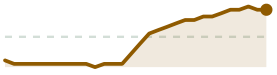
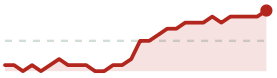
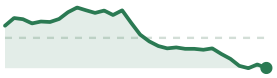
Suresh Pal — 0 Stable
59 · Nighasan, Lakhimpur Kheri
HR 75 BP 141/90 SpO_2 96 36.7° 150

Mohd. Irfan

55 · Sandila, Hardoi · COPD

High risk | Source: Wearable plus pulse oximeter

10
risk score · last reading 3:22 pm

Heart rate baseline 84	**126** bpm		+2
Blood pressure baseline 124/78	**108/66** mmHg		+1
SpO_2 baseline 95	**87** %		+3
Temperature baseline 36.9	**38.5** °C		+1
Glucose baseline 104	**106** mg/dL		ok
Respiratory rate baseline 18	**27** breaths/min		+3
Activity level baseline 200 (not scored)	**31** steps/hour		ok

AI EARLY-WARNING FORECAST

Already high risk. Escalation is active.

Pulse +2 | BP +1 | SpO_2 +3 | Temp +1 | Breathing +3

Bars show which vitals add to the score. The demo forecast is a linear projection of the last 6 scores. A production version would use a trained time-series model validated on clinical data.

STATISTICAL ANOMALY CHECK

Unusual compared with this patient's own baseline (χ^2 = 163.1, 7 df, p < 0.001)

Vital	Now	Baseline	z-score	EWMA
Pulse	126	84	+7.0	125

Systolic BP	108	124	-2.0	109
SpO_2	87	95	**-6.7**	87
Temperature	38.5	36.9	**+6.4**	38.5
Glucose	106	104	+0.2	107
Breathing rate	27	18	**+4.5**	26
Activity (steps/hr)	31	200	-2.1	48

z = (reading minus personal baseline) divided by typical day-to-day variation. EWMA (λ = 0.3) smooths noise. χ^2 adds the squared z-scores and assumes independent vitals; a production model would use a full covariance (Mahalanobis) estimate.

AI CLINICAL COPILOT (LLM)

An LLM reads the vitals, z-scores and forecast above, then drafts an SBAR note, questions to ask, and a Hindi advisory. It supports the doctor and never replaces them.

✨ Generate note with AI

FOLLOW-UP

Suggested contact: within 1 hour. No appointment booked yet.

2:42 pm Auto-escalated to ASHA worker and 108 advisory (no doctor response in 15 min)

TREND SUMMARY

Over the last 12 readings (55 min), heart rate rose from 116 bpm to 126 bpm; SpO_2 fell from 90% to 87%; temperature rose from 38°C to 38.5°C. Risk score moved from 7 to 10 (High risk). Against this patient's own baseline: pulse +42, systolic BP −16. Suggested: call the patient or ASHA worker now and offer an eSanjeevani teleconsult.

Written from rules in this demo. A production version would draft this with a language model and keep the doctor's review step.

Request follow-up call | Notify ASHA worker | Start eSanjeevani teleconsult | Simulate recovery

FOLLOW-UP LOG

2:42 pm Auto-escalated to ASHA worker and 108 advisory (no doctor response in 15 min)

EARLY-WARNING ALERTS

High risk **Savitri Devi** 3:02 pm · App + SMS

SMS IN ENGLISH

ArogyaWatch: Savitri Devi, 72. HIGH RISK (score 7). Sugar 305, Breathing 22/min. Please review now.

SMS IN HINDI

आरोग्यवॉच: सावित्री देवी, 72 वर्ष। अधिक जोखिम (स्कोर 7)। Sugar 305, Breathing 22/min। कृपया तुरंत देखें।

Confirm | False alarm | Auto-escalated: ASHA + 108

Needs review **Savitri Devi** 2:47 pm · App + SMS

SMS IN ENGLISH
ArogyaWatch: Savitri Devi, 72. Needs review (score 6). Sugar 292, Breathing 21/min. Please review today.

SMS IN HINDI
आरोग्यवॉच: सावित्री देवी, 72 वर्ष। समीक्षा ज़रूरी (स्कोर 6)। Sugar 292, Breathing 21/min। कृपया आज देखें।

Confirm | False alarm

High risk **Mohd. Irfan** 2:27 pm · App + SMS

SMS IN ENGLISH
ArogyaWatch: Mohd. Irfan, 55. HIGH RISK (score 7). SpO_2 90%, HR 116. Please review now.

SMS IN HINDI
आरोग्यवॉच: मोहम्मद इरफ़ान, 55 वर्ष। अधिक जोखिम (स्कोर 7)। SpO_2 90%, HR 116। कृपया तुरंत देखें।

Confirm | False alarm | Auto-escalated: ASHA + 108

Needs review **Savitri Devi** 2:27 pm · App + SMS

SMS IN ENGLISH
ArogyaWatch: Savitri Devi, 72. Needs review (score 5). Sugar 268, Breathing 21/min. Please review today.

SMS IN HINDI
आरोग्यवॉच: सावित्री देवी, 72 वर्ष। समीक्षा ज़रूरी (स्कोर 5)। Sugar 268, Breathing 21/min। कृपया आज देखें।

Confirm | False alarm

Needs review **Mohd. Irfan** 2:17 pm · App + SMS

SMS IN ENGLISH
ArogyaWatch: Mohd. Irfan, 55. Needs review (score 6). SpO_2 91%, Breathing 22/min. Please review today.

SMS IN HINDI
आरोग्यवॉच: मोहम्मद इरफ़ान, 55 वर्ष। समीक्षा ज़रूरी (स्कोर 6)। SpO_2 91%, Breathing 22/min। कृपया आज देखें।

Confirm | False alarm

DATA PIPELINE AND AI BRIEFING

Wearables ×4 | IoT sensors ×2 | Simulated dataset ×7 | Manual entry ×1 →

Edge gateway: offline cache + SMS → Cloud API → NEWS2 rules + z-score + forecast →

Dashboard, alerts, LLM summaries

✨ AI shift briefing for the doctor

The briefing ranks patients and suggests actions. Doctors make the clinical decision.

DATA SCIENCE CHECK: SIMULATED COHORT EVALUATION

Generates 600 synthetic patients (half stable, half deteriorating) with the same simulator, then compares a plain threshold alert (score 7 or more) with the forecast early warning. Synthetic data only: it demonstrates the method, not clinical accuracy.

Run 600-patient simulation

All patients, vitals and alerts here are simulated for the demo. Scores follow the NEWS2 bands for pulse, systolic BP, SpO_2 (scale 1), temperature and respiratory rate. Consciousness and supplemental oxygen are not collected, activity level is tracked but not scored, and the glucose bands are demo thresholds set by the team that need clinical validation. The trend summary is generated from rules in this prototype. This is not medical advice.